

Hybrid CNN-VGG19 Approach for Real-Time Contactless Fingerprint Recognition

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Abstract— Fingerprint recognition is essential in biometric authentication systems which provides a secure identity verification. Traditional fingerprint sensors face issues such as wear, inconsistent pressure and hygiene issues caused by physical contact. These limitations along with environmental factors like lighting and angle inconsistencies can degrade image quality that affect recognition accuracy. To address these challenges, this work focuses on contactless approach integrated with deep learning. High-resolution cameras or mobile devices capture fingerprint images offering a hygienic and efficient solution. Hybrid CNN-VGG19 -based framework enhances images through noise reduction, distortion correction and resolution enhancement ensuring the accurate extraction of fingerprint patterns like minutiae points and ridges. This model minimizes loss over multiple epochs with metrics like accuracy, precision and F1 score to evaluate real-world performance. This approach prevents overfitting thus ensuring reliable contactless fingerprint recognition system for modern applications. Its main objective is to achieve real-time processing authentication meeting the demands of high-throughput environments. By using the latest advancements in computer vision and deep learning, this solution sets a new benchmark in technology development.

Keywords: Contactless fingerprint, Deep learning, hybrid approach, CNN, VGG19, Image enhancement, Biometric systems.

I. INTRODUCTION

Contactless fingerprint recognition has become a revolutionary advancement in biometric authentication, addressing the limitations of traditional fingerprint scanners. Conventional systems require direct physical contact, leading to concerns such as device wear, inconsistent pressure application. The shift toward contactless recognition eliminates these issues by leveraging high-resolution imaging and deep learning techniques. Among the various approaches, the Hybrid CNN-VGG19 model stands out for its ability to deliver high accuracy, efficiency, and adaptability in real-time applications. This hybrid approach integrates Convolutional Neural Networks (CNNs) with VGG19, a deep learning model known for its robust feature extraction capabilities, making it highly suitable for analyzing fingerprint images in a contactless environment.

The process starts with high-resolution cameras or mobile devices capturing fingerprint images, eliminating the need for physical touch. However, contactless imaging presents challenges such as variations in lighting, perspective distortion, and background noise, which can affect recognition performance. To overcome these challenges, preprocessing techniques such as noise reduction, distortion correction, and

contrast enhancement are applied to standardize the images before feeding them into the hybrid model. These preprocessing steps ensure that the fingerprint features remain clear and distinguishable, regardless of environmental conditions.

In this model, CNNs play a fundamental role in extracting spatial features such as ridges, minutiae points, and fingerprint patterns. These features are then processed by VGG19, which refines and enhances them through its 19-layer deep architecture. VGG19, pre-trained on large-scale datasets, enhances the feature extraction process by recognizing fine details in fingerprint images. The extracted features are then passed to a fully connected layer, where classification is performed to map each fingerprint to a unique identity. By combining CNNs' spatial feature extraction ability with VGG19's deep learning strength, the hybrid model ensures high recognition accuracy and computational efficiency.

A key advantage of this approach is its real-time performance, making it highly suitable for applications requiring fast and accurate authentication. This work is optimized for speed and scalability, enabling it to be used in critical applications such as Aadhaar-based authentication, national entrance exams, and secure access control systems. To further enhance efficiency, techniques such as model pruning, quantization, and deployment on edge AI devices are implemented, ensuring that fingerprint recognition is completed within milliseconds. This optimization enables the system to operate seamlessly in real-world environments without compromising accuracy.

The model's performance is evaluated using metrics such as Precision, Recall, F1-Score, and Equal Error Rate (EER), which assess its reliability and robustness. The results demonstrate high accuracy and consistency, even under challenging conditions such as poor lighting or image distortions. The system effectively adapts to different scenarios, ensuring stable and secure authentication.

II. RELATED WORKS

MCLFIQ (Mobile Contactless Fingerprint Image Quality) refers to a system that evaluates the quality of fingerprint images captured using mobile phones or standard cameras without physical contact. It focuses on assessing factors such as clarity, noise, distortion, and resolution, ensuring reliable authentication. MCLFIQ plays a crucial role in enhancing biometric systems for secure, hygienic, and

accurate identification. The system utilizes deep learning and image processing techniques to improve fingerprint image quality in various conditions. Its goal is to overcome challenges like smudges, inconsistent pressure, and environmental variations in image capture[1].

A system for matching contactless fingerprints to contact-based impressions, using Deep Print and Veri Finger for feature extraction, achieving high True Acceptance Rates (TARs) and robust performance despite distortions and varying quality [9]. A tool for assessing fingerprint image quality, improving reliability and interoperability with better quality scores and lower computational complexity, enhancing mobile platform support [10]. A deep learning-based approach using DeepLabv3+ to segment fingertips from 2D images, achieving improved performance for touchless fingerprint recognition with a high Intersection over Union (IoU) score [11]. An approach to using smartphone cameras for fingerprint quality assessment, extracting features from ridge pattern autocorrelation in both spatial and DCT domains, demonstrating practical applicability [12]. Explores the evolution of contactless systems, addressing challenges like hygiene and elastic deformation while focusing on improved image quality and feature extraction for large-scale deployment [13]. Discusses challenges of matching contactless fingerprints to traditional systems due to distortions, highlighting the need for research to improve accuracy and performance [14].

A deep learning approach that generates fixed-length representations of fingerprints for efficient and secure identification, achieving strong performance even with varying fingerprint quality [15]. Emphasizes the rise of touchless systems, highlighting the advantages of improved hygiene and innovative methods like monogenic-wavelet-based algorithms for minutiae extraction [16]. A method using ridge orientation profiles and invariant features for robust biometric verification, integrating data fusion across finger segments [17]. A multi-Siamese CNN architecture for robust contactless-to-contact fingerprint comparisons, leveraging minutiae and ridge maps [18]. Development of a 3D scanner with a new segmentation method, improving the accuracy of 3D fingerprint recognition [19]. A method for biometric authentication using smartphone cameras, showing promising accuracy despite device variability [20]. A no-reference image quality assessment model outperforming PSNR and SSIM, ideal for real-time applications [21].: A smartphone-based system for fingerprint recognition that enhances performance across diverse conditions with effective segmentation and enhancement [22]. A method for fingertip detection that performs well even in closed hand scenarios, enhancing gesture recognition [23]. A method that classifies fingerprint quality using features based on ridge patterns' spatial autocorrelation, demonstrating its effectiveness [24]

Discusses the persistent challenges in touchless systems such as low contrast and 3D to 2D image mapping, suggesting solutions to optimize image quality and improve matching accuracy [25]. These works contribute to the development of more accurate and scalable contactless

fingerprint recognition and image quality assessment technologies.

III. METHODOLOGY

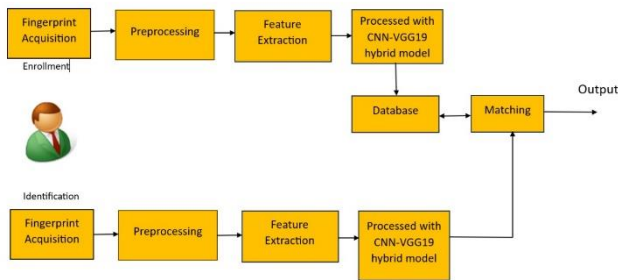
The Hybrid CNN-VGG19 approach is a deep learning-based methodology for real-time contactless fingerprint recognition. This approach addresses the limitations of traditional fingerprint scanners by using mobile cameras or optical sensors to capture fingerprint images. Unlike conventional systems that rely on direct sensor contact, this method ensures hygienic, efficient, and accurate fingerprint authentication. The methodology involves multiple phases, including fingerprint acquisition, preprocessing, feature extraction, model processing, database storage, matching, and decision-making. Each of these steps is crucial to achieving high accuracy and reliability in fingerprint authentication.

a) **ENROLLMENT PHASE:** A user's fingerprint is captured and stored in the database for future authentication. The process begins with fingerprint acquisition, where the image is taken using a high-resolution camera. Since mobile cameras introduce variations due to lighting conditions, angle distortions, and background noise, preprocessing is essential. This stage refines the image quality using several techniques. Noise removal is performed using Gaussian and median filters to eliminate unwanted distortions. Contrast enhancement is applied using histogram equalization to improve the visibility of fingerprint ridges and minutiae points. Additionally, geometric alignment ensures that all fingerprints are consistently oriented before feature extraction. Finally, segmentation is used to isolate the fingerprint region from the background, removing any unnecessary information that might interfere with feature extraction. After preprocessing, feature extraction is performed using a hybrid approach that integrates CNN (Convolutional Neural Networks) and VGG19 (a deep learning model with 19 layers). The CNN extracts fundamental spatial features, such as ridge patterns, while VGG19 refines these features by learning complex fingerprint structures. The combined power of these two models enhances feature representation, improving accuracy. The extracted feature vectors are stored as templates in a secure fingerprint database, ready for future identification. The entire enrollment process ensures that each fingerprint is properly processed and stored in a format that allows for seamless matching during authentication.

b) **IDENTIFICATION PHASE:** The system compares a newly captured fingerprint with the stored templates to verify the user's identity. This phase starts with **fingerprint** acquisition, where the user presents their fingerprint for recognition. Given that environmental factors such as lighting conditions and background noise can introduce variations, preprocessing is once again applied to refine the image. The preprocessing techniques remain the same—noise removal, contrast enhancement, geometric alignment, and segmentation—ensuring that the newly captured fingerprint matches the format of the enrolled fingerprints. Once preprocessing is complete, feature extraction is conducted using CNN and VGG19 to maintain consistency with the stored fingerprint templates. The

extracted feature set is converted into a numerical representation, which is crucial for efficient comparison. The Hybrid CNN-VGG19 model then processes the extracted features and transforms them into a format suitable for matching. The model applies deep learning techniques to analyze fingerprint ridge structures, minutiae points, and texture patterns, ensuring high accuracy. To determine whether the new fingerprint matches any stored templates, the system employs fingerprint matching algorithms such as Euclidean distance and cosine similarity. These techniques measure the similarity between the extracted feature vector of the input fingerprint and the stored templates. If the similarity score falls within an acceptable threshold, authentication is successful, and the user is granted access. If the similarity score is below the threshold, authentication fails, and the user may be denied access or prompted to recapture their fingerprint.

c) **DECISION MAKING PHASE:** The decision-making process is the final step in fingerprint recognition. If a match is found, the system grants access, confirming the user's identity. If no match is found, access is denied, and the user may be required to reattempt authentication. The decision-making logic ensures a **balance between security and usability**, reducing false positives and false negatives.



Block diagram of the hybrid CNN-VGG19 approach

The Hybrid CNN-VGG19 approach stands out for its robustness against environmental variations, overcoming challenges faced by traditional fingerprint scanners, such as inconsistent pressure application, smudges, and sensor degradation. By ensuring clear and consistent fingerprint capture, this contactless method enhances usability across various domains, including biometric security, financial transactions, and national identity verification. Moreover, the integration of deep learning models, advanced preprocessing techniques, and real-time optimizations makes this approach highly accurate, secure, and scalable for real-time fingerprint recognition. Its hygienic, efficient, and reliable authentication process offers a practical alternative to conventional scanners, making it particularly suitable for applications in smartphones, banking, national identity systems, and access control. The ability to function effectively in real-world conditions, optimize performance for edge devices, and maintain high accuracy highlights the significant advancements this approach brings to fingerprint recognition technology.

IV. EVALUTION

The evaluation of the **Hybrid CNN-VGG19 approach** for contactless fingerprint recognition demonstrates strong performance with high accuracy, precision, and recall. The accuracy graph will show a steady improvement, with both training and validation accuracy stabilizing close indicating effective learning and generalization. The loss graph highlights a consistent decline in both training and validation loss, confirming that the model minimizes errors without significant overfitting. The classification report reveals that most fingerprint samples achieve excellent recognition, with high precision, recall, and F1-scores, demonstrating the model's robustness.

V. PERFORMANCE COMPARSION

The Table 5.1 represents a comparative analysis of different machine learning and deep learning algorithms based on four evaluation metrics: Precision, Recall, F1-score, and Accuracy. Among the traditional machine learning models, Random Forest (RF) achieved an accuracy of 80.03%, while Support Vector Machine (SVM) slightly underperformed with 78.96%. Deep learning models performed significantly better, with VGG19 achieving an accuracy of 85.19% and ResNet101 further improving it to 88.26%. The highest performance was achieved by the Hybrid CNN-VGG19 model, which outperformed all others with an impressive accuracy of 95.63%. This hybrid approach also recorded the highest precision (91.99%), recall (94.95%), and F1-score (95.93%), indicating its superior capability in handling the given dataset. The results suggest that deep learning models, especially hybrid architectures, are more effective in achieving higher classification performance compared to traditional machine learning techniques.

Table 5.1 Comparison Table

SL.NO	ALGORITHM USED	PRECISION	RECALL	F1-score	ACCURACY
1	RF	80.03	83.65	80.81	80.03
2	Svm	78.86	79.63	78.76	78.96
3	Vgg19	85.12	83.12	85.72	85.19
4	ResNet 101	88.09	86.78	88.98	88.26
5	Hybrid cnn-vgg19	91.99	94.95	95.93	95.63

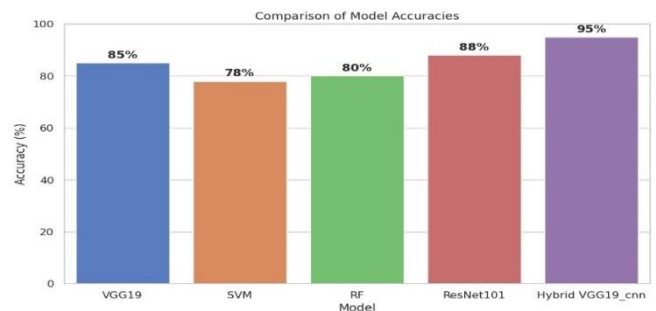


Fig 5.1 Comparison chart of accuracy

The bar chart Fig 5.1 compares the accuracy of different machine learning and deep learning models. Traditional models like SVM (78%) and RF (80%) performed moderately, while deep learning models—VGG19 (85%)

From Fig 5.3 and ResNet101 (88%) from Fig 5.2 showed better accuracy. The Hybrid VGG19-CNN model achieved the highest accuracy at 95%, demonstrating the superiority of hybrid deep learning architectures.

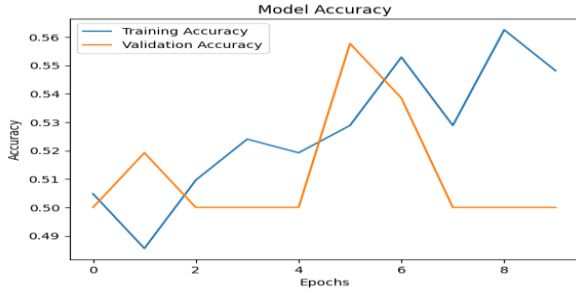


Fig 5.2 Accuracy of ResNet 101

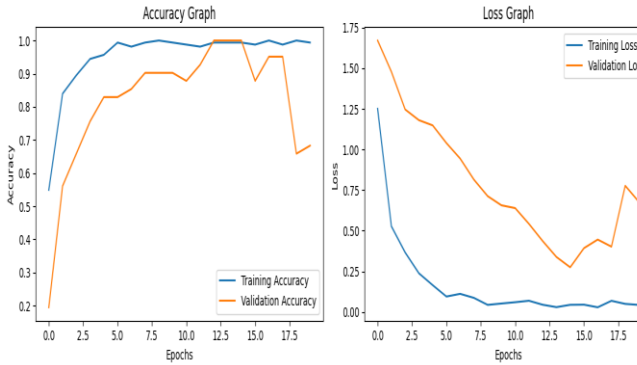


Fig 5.3 Accuracy of VGG19

VI. RESULT AND DISCUSSION

In conclusion, the evaluation of the Hybrid CNN-VGG19 approach demonstrates its suitability for high-quality contactless fingerprint recognition. Its high accuracy, precision, recall, and F1-scores reflect its potential to improve security and convenience in authentication systems, making it a valuable solution in various biometric applications. The model's ability to handle variations in image quality, such as distortion and noise, further enhances its reliability in real-world scenarios.

Additionally, its adaptability to different datasets showcases its scalability for diverse fingerprint recognition systems. Further improvements, such as optimization for real-time performance and integration with mobile devices, could make this approach even more accessible and effective in the field of biometric authentication. Moreover, integrating with advanced techniques like transfer learning and fine-tuning could further refine the model's performance. The use of lightweight architectures could also enhance deployment efficiency, particularly in resource-constrained environments. With continued advancements, this hybrid approach has the potential to revolutionize fingerprint-based security, offering a seamless and secure authentication experience.

VII. CONCLUSION

This effectively combines the strengths of Convolutional Neural Networks (CNN) and the VGG19 architecture to address the challenges associated with real-time, contactless fingerprint identification. By leveraging the robust feature extraction capabilities of VGG19 alongside the adaptability and power of CNN, the system significantly enhances the accuracy and reliability of fingerprint recognition in real-world scenarios, even when dealing with low-resolution or distorted images.

This hybrid approach overcomes many limitations of traditional fingerprint recognition systems, such as issues caused by sensor quality, variations in image quality due to environmental factors, and physical distortions in captured prints. The model's real-time processing capability further contributes to its practical applicability in various security-critical systems, ensuring a more efficient, hygienic, and user-friendly solution for biometric authentication. The results demonstrate that this approach not only improves fingerprint classification accuracy but also maintains high performance even in less-than-ideal conditions. The combination of CNN and VGG19 paves the way for future advancements in contactless biometric authentication, with potential applications across various fields such as security systems, mobile applications, and identity verification processes.

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